3.Programming security and permissions

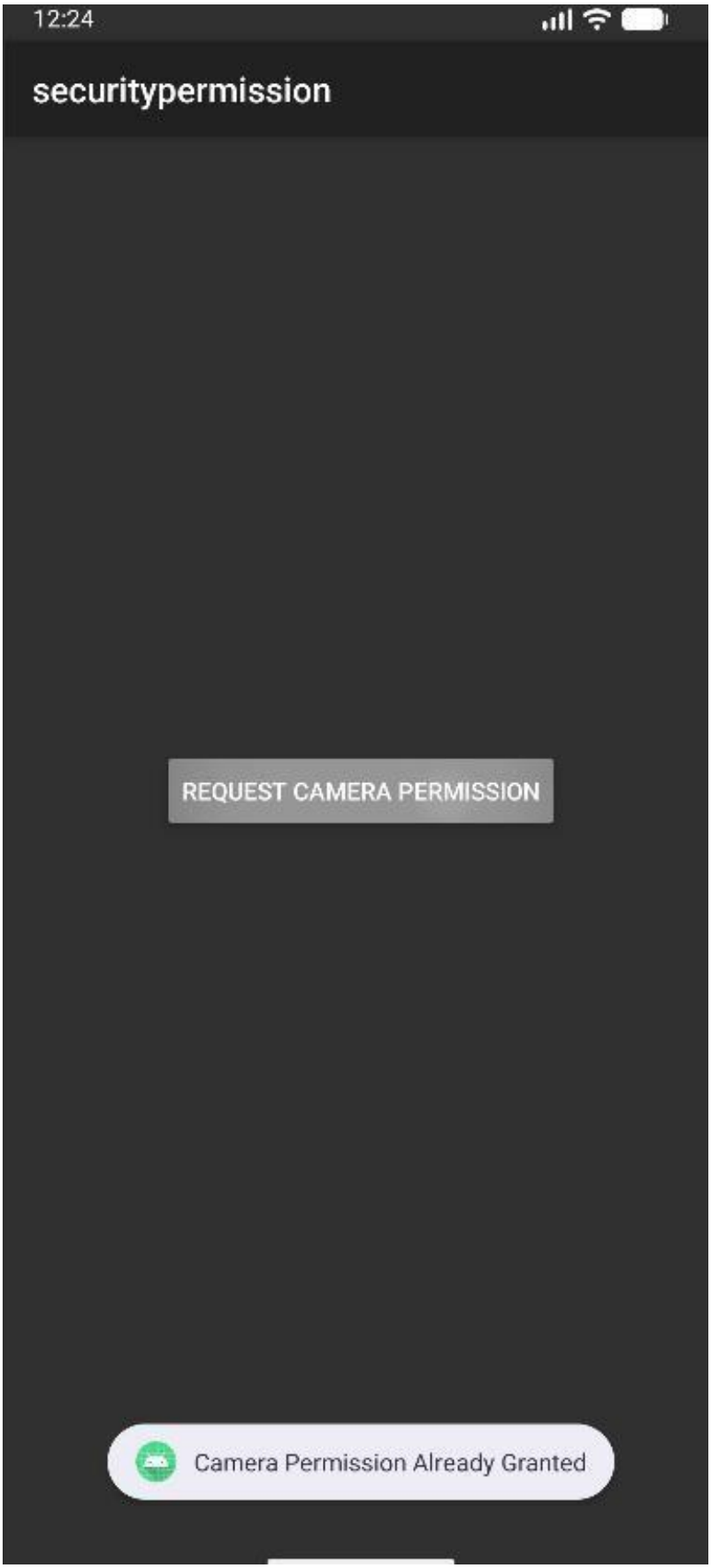
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <Button
        android:id="@+id/btnPermission"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Request Camera Permission"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.securitypermission;
import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
public class MainActivity extends AppCompatActivity {
    Button btnPermission;
    private static final int CAMERA_PERMISSION_CODE = 100;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnPermission = findViewById(R.id.btnPermission);
        btnPermission.setOnClickListener(v -> checkPermission());
    }
    private void checkPermission() {
        if (ContextCompat.checkSelfPermission(this, Manifest.permission.CAMERA)
                == PackageManager.PERMISSION_GRANTED) {
            Toast.makeText(this, "Camera Permission Already Granted",
Toast.LENGTH_SHORT).show();
        } else {
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.CAMERA},
                    CAMERA_PERMISSION_CODE);
        }   }
    @Override
    public void onRequestPermissionsResult(int requestCode, String[] permissions, int[]
grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == CAMERA_PERMISSION_CODE) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                Toast.makeText(this, "Camera Permission Granted",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Camera Permission Denied",
Toast.LENGTH_SHORT).show();
            }
    }    } }
```


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securitypermission
REQUEST CAMERA PERMISSION
Camera Permission Already Granted